25. Ordered Rings and Fields

1. Under $P_{\text{high}}, a - 0$ is positive, $x - a$ is positive, $x^2 - x$ is positive, etc. Thus we have the ordering $a < x < x^2 < x^3 < \cdots < x^n < \cdots$ for any $a \in R$ and $n \in \mathbb{Z}^+$.

2. The ordering is P_{low} so $(x^i - x^j) \in P$ and $x^i > x^j$ if $i < j$. Thus we have the ordering $\cdots < x^3 < x^2 < x < x^0 = 1 < x^{-1} < x^{-2} < x^{-3} < \cdots$.

3. Because $\sqrt{2}$ is negative, we must have $n < 0$ for $n\sqrt{2}$ to be positive. We see that $m + n\sqrt{2}$ is positive if $m > 0$ and $n < 0$, or if $m > 0$ and $m^2 > 2n^2$, or if $n < 0$ and $2n^2 > m^2$.

4. (i) $a\ c\ d\ e\ b$ (ii) $d\ b\ a\ e\ c$ **5.** (i) $a\ c\ e\ d\ b$ (ii) $e\ c\ b\ a\ d$

6. (i) $c\ a\ b\ e\ d$ (ii) $e\ c\ a\ b\ d$ **7.** (i) $d\ a\ b\ c\ e$ (ii) $d\ c\ e\ a\ b$

8. (i) $e\ a\ c\ b\ d$ (ii) $c\ d\ a\ e\ b$ **9.** (i) $c\ a\ e\ d\ b$ (ii) $e\ c\ b\ a\ d$

10. $b\ d\ e\ a\ c$ **11.** $d\ b\ a\ e\ c$ **12.** $b\ e\ c\ d\ a$ **13.** $d\ e\ b\ a\ c$

14. The smallest subfield of $\mathbb{C}$ containing $\sqrt[3]{2}$ is the intersection F of all subfields of $\mathbb{C}$ that contain $\sqrt[3]{2}$. Because $\mathbb{R}$ is one such subfield of $\mathbb{C}$, we see that $F \leq \mathbb{R}$ and thus can be ordered using the induced ordering from $\mathbb{R}$. Because the smallest subfield K of $\mathbb{C}$ containing $\alpha = \sqrt[3]{2}(\frac{-1+i\sqrt{3}}{2})$ is isomorphic to F, Theorem 25.10 shows that K can be ordered, and K contains α which is not a real number.

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16. Let $\mathbb{Q}[x]$ be ordered using the ordering P_{high}, in which x is greater than every element of $\mathbb{Q}$. Let $\mathbb{Q}[\pi]$ have the ordering provided by the ordering P_{high} of $\mathbb{Q}[x]$ and the isomorphism of $\mathbb{Q}[x]$ with $\mathbb{Q}[\pi]$ provided by the evaluation homomorphism $\phi_\pi : \mathbb{Q}[x] \to \mathbb{R}$, as described in Example 25.11. Because $\phi_\pi(x) = \pi$ and $\phi_\pi(q) = q$ for all $q \in \mathbb{Q}$, the inequality $q < x$ in $\mathbb{Q}[x]$ is carried into $q < \pi$ in $\mathbb{Q}[\pi]$ for all $q \in \mathbb{Q}$.

17. Let $m + n\sqrt{2}$ and $r + s\sqrt{2}$ be any elements of $\mathbb{Z}[\sqrt{2}]$, so that $m, n, r, s \in \mathbb{Z}$. Then

$$\begin{aligned}\phi((m + n\sqrt{2}) + (r + s\sqrt{2})) &= \phi((m + r) + (n + s)\sqrt{2}) \\ &= (m + r) - (n + s)\sqrt{2} \\ &= (m - n\sqrt{2}) + (r - s\sqrt{2}) \\ &= \phi(m + n\sqrt{2}) + \phi(r + s\sqrt{2}),\end{aligned}$$

showing that ϕ is an additive homomorphism. Turning to the multiplication, we have

$$\begin{aligned}\phi((m + n\sqrt{2})(r + s\sqrt{2})) &= \phi((mr + 2ns) + (ms + nr)\sqrt{2}) \\ &= (mr + 2ns) - (ms + nr)\sqrt{2} \\ &= (m - n\sqrt{2})(r - s\sqrt{2}) \\ &= \phi(m + n\sqrt{2})\phi(r + s\sqrt{2}).\end{aligned}$$

Thus we have a homomorphism. It is clear that ϕ is a one-to-one map of $\mathbb{Z}(\sqrt{2})$ onto $\mathbb{Z}(\sqrt{2})$, so ϕ is an isomorphism.

18. If $a \in P$, then $a - 0 = a \in P$. By definition of $<$ in Theorem 25.5, $(a - 0) \in P$ means that $0 < a$.

19. If $c = 0$, then $ac = bd = 0$. Because $b \in P$ and R can have can no divisors of zero, we conclude that $d = 0$. By similar argument, $d = 0$ implies that $c = 0$.

Suppose now that c and d are nonzero. From $ac = bd$, we obtain $acd = bd^2$. Now $b \in P$, and $d^2 \in P$ implies $bd^2 \in P$ so $acd \in P$. Then $-acd = a(-cd) \notin P$, so $-cd \notin P$ and thus $cd \in P$.

20. If $a < b$, then $(b - a) \in P$. Now $b - a = (-a) - (-b)$ so $((-a) - (-b)) \in P$. Thus $-b < -a$.

21. If $a < 0$ then $(0 - a) \in P$ so $-a \in P$. If $0 < b$ then $(b - 0) \in P$ so $b \in P$. Consequently $(-a)b = -(ab) \in P$. Thus $(0 - ab) \in P$ so $ab < 0$.

22. Either $a/b \in P$ or $-(a/b) \in P$. If $-(a/b) \in P$, then $b(-a/b) = -a \in P$, contradicting the hypothesis that $a \in P$, so $a/b \in P$.

23. Now $0 < a$ implies that $a = (a - 0) \in P$, and $a < 1$ implies that $(1 - a) \in P$. Either $(\frac{1}{a} - 1) \in P$ or $(1 - \frac{1}{a}) \in P$. If $(1 - \frac{1}{a}) \in P$, then $(a - 1) = a(1 - \frac{1}{a}) \in P$, contradicting $(1 - a) \in P$. Thus $(\frac{1}{a} - 1) \in P$ so $1 < \frac{1}{a}$.

24. Now $-1 < a$ implies that $(a + 1) \in P$ and $a < 0$ implies that $(0 - a) = -a \in P$. If $(\frac{1}{a} + 1) \in P$, then $(-1 - a) = -a(\frac{1}{a} + 1)$ is in P, contradicting $(a + 1) \in P$. Therefore $-(\frac{1}{a} + 1) = (-1 - \frac{1}{a}) \in P$, so $\frac{1}{a} < -1$.

25. *Closure:* Let $a', b' \in P'$ and let $\phi(a) = a'$ and $\phi(b) = b'$. Because ϕ is one to one and $P' = \phi[P]$, we must have $a \in P$ and $b \in P$. Therefore $ab \in P$, so $\phi(ab) = \phi(a)\phi(b) = a'b' \in P'$. Likewise $(a + b) \in P$ so $\phi(a + b) = \phi(a) + \phi(b) = (a' + b') \in P$.

Trichotomy: Let $c' \in R'$ and let c be the unique element of R such that $\phi(c) = c'$. If $c \in P$, then $\phi(c) = c' \in P$. If $c = 0$, then $\phi(c) = c' = 0'$. If $-c \in P$, then $\phi(-c) = -\phi(c) = -c' \in P'$. The fact that only one of $c \in P, c = 0, -c \in P$ holds shows that only one of $c' \in P', c' = 0', -c' \in P'$ holds.

Furthermore, $a < b$ in R if and only if $(b - a) \in P$, which is true if and only if $\phi(b - a) = (\phi(b) - \phi(a)) \in P'$, which is true if and only if $\phi(a) <' \phi(b)$.

26. *Closure:* Let $a, b \in P \cap S$. Then $ab \in P$ by closure of P and $ab \in S$ by closure of S as a subring. Thus $ab \in P \cap S$. Likewise, $(a + b) \in P$ and $(a + b) \in S$ so $(a + b) \in P \cap S$.

Trichotomy: Let $s \in S$. Then $s \in R$ so either $s = 0, s \in P$, or $-s \in P$, and only one of these holds. Thus in S, either $s = 0$, $s \in S \cap P$, or $-s \in S \cap P$, and only one of these holds.

27. Let $<$ be a relation on R satisfying trichotomy, transitivity, and isotonicity as stated in Theorem 25.5. Let $P = \{x \in R \mid 0 < x\}$.

Closure: Let $x, y \in P$. Then $0 < x$ and $0 < y$. By the second condition in isotonicity, we have $0y < xy$ so $0 < xy$ and $xy \in P$. Also, from $0 < x$ and the first condition of isotonicity, we obtain $0 + y < x + y$, so $y < x + y$. From $0 < y$ and $y < x + y$, we obtain $0 < x + y$ by transitivity.

Trichotomy for P: Let $x \in R$. By trichotomy for $<$, precisely one of $0 < x, 0 = x$, or $x < 0$ holds. Now $0 < x$ if and only if $x \in P$. By isotonicity, $x < 0$ implies $(-x + x) < (-x + 0)$. Again by isotonicity, $(-x + x) < (-x + 0)$ implies $x + (-x + x) < x + (-x + 0)$, that is, it implies $x < 0$. Thus $x < 0$ if and only if $0 < -x$, which is true if and only if $-x \in P$.

By isotonicity, $a < b$ for the given $<$ on R if and only if $-a + a < -a + b$, that is, if and only if $0 < b - a$. By definition of $P, 0 < b - a$ if and only if $(b - a) \in P$, which is true if and only if $a <_P b$. Thus $<$ and $<_P$ are the same relation on R.